



RISK MANAGEMENT

Table of Contents

1.Part 1: Applying the Select and Focus Phase to Project Stardust

- 1.1 Introduction
- 1.2 Project Context: Execution and Delivery in Stardust
- 1.3 Identifying and Prioritizing Key Uncertainties in Execution
- 1.4 Tailoring the Select and Focus Phase to Project Stardust
- 1.5 Overcoming Challenges in Adaptation of Select and Focus for Stardust
- 1.6 Conclusion

2.Part 2: Shaping Base Plans Using Models – Project Stardust

- 2.1 Introduction to the Shape Base Plans Phase
- 2.2 Modelling Stardust's Execution Pathway
- 2.3 Structure of the Base Plan Model
 - 2.3.1 Task Breakdown
 - 2.3.2 Time Parameters and Slack Analysis
 - 2.3.3 Network Diagram
 - 2.3.4 Gantt Chart
- 2.4 Incorporating Uncertainty in the Model
- 2.5 Clarity and Uncertainty
- 2.6 Aspirational vs. Commitment Targets
- 2.7 Using the Model for Decision-Making and Iteration
- 2.8 Feedback Loops and Organizational Learning
- 2.9 Conclusion

3.Part 3: PMBOK® Guide Limitations in Managing Project Stardust Risk

- 3.1 Introduction
- 3.2 Overemphasis on Process and Control
- 3.3 Lack of Distinction Between Risk and Uncertainty
- 3.4 Inflexibility to Developing Clarity
- 3.5 Weak Support for Opportunity Efficiency
- 3.6 Stakeholder Engagement Treated as Peripheral

3.7 Insufficient Emphasis on Modelling and Iteration

3.8 Conclusion

4.References

Part 1: Applying the Select and Focus Phase to Project Stardust

1. Introduction: The Role of the Select and Focus Phase in Enlightened Planning

The Select and Focus is the beginning step of the Specific Process for Projects (SPP) in the Enlightened Planning framework. It departs from risk to selecting key uncertainties where additional clarity will add value to decisions—an opportunity efficiency strategy (Brito & Stafford). Instead of rigid tools like probability-impact matrices, it promotes dynamic learning and adaptation. For an intricate project such as Stardust, this step helps to harmonize stakeholders, eliminate uncertainties, and prepare for implementation in a structured but flexible way.

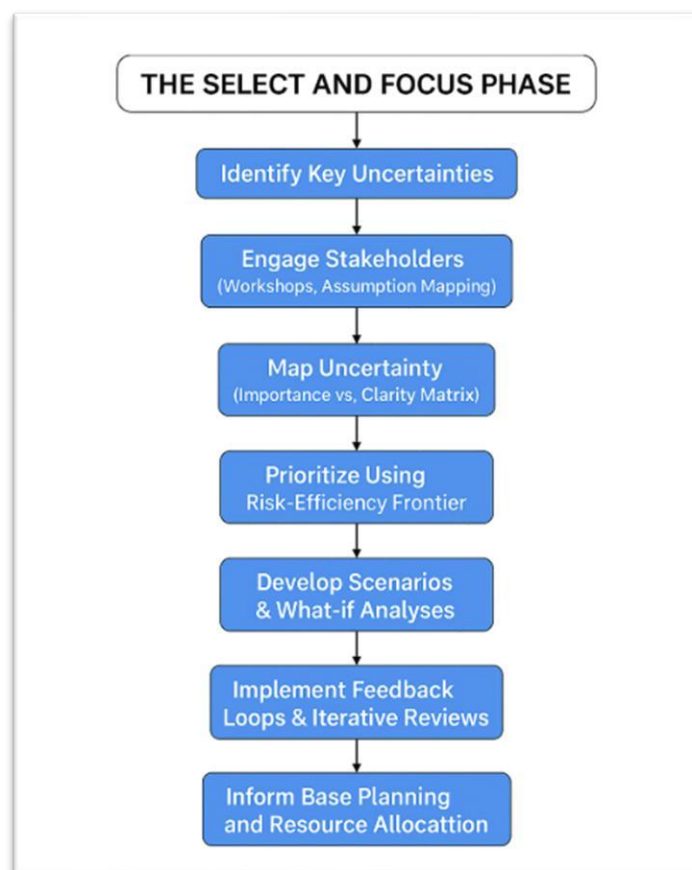


Fig 1. Select and Focus Phase

2. Project Context: Execution and Delivery in Project Stardust

Project Stardust launched by Stellar Innovations, is designed to transform space component manufacturing through robotics, 3D printing, and advanced materials with a focus on scalability, speed, and sustainability. As the project moves from R&D to Execution and Delivery, uncertainties rise with complex systems, regulatory control, and high reliability requirements. Applying the Select and Focus phase helps the team choose and prioritize critical uncertainties to safeguard timelines, integration, and stakeholder expectations.

3. Identifying and Prioritizing Key Uncertainties in Execution

Select and Focus filters out the unknowns to eliminate high impact or low clarity uncertainties.

Project Stardust's core uncertainties are as follows:

- a. **Material Performance:** Space-bound untested alloy materials and composites where realtime validation cannot occur make it a core unknown.
- b. **Reliability of Robotized Manufacturing:** Robotized production lines may go without flaw when done at minimal levels but carry with them possibilities of cascading failure and flaw at scale levels.
- c. **Certification Timelines:** Taut and evolving regulation processes cause delays and cost inflation, especially with design changes.
- d. **Supply Chain Volatility:** Geopolitical tensions and dependency on rare-earth elements threat production continuity.
- e. **Stakeholder Alignment:** Conflicting expectations of customer, regulator, and internal objectives can cause scope changes and damage to reputation.

4. Tailoring the Select and Focus Phase to Project Stardust

To fit Select and Focus of SSP to Project Stardust, one needs a very structured understanding of the execution environment. The 7Ws framework -**What, Why, Where, When, Who, Which Way, and How Much** serves to provide the broader view through which this tailoring can be presented, regarding the innovative and multi-stakeholder perspective of Project Stardust.

What?

Within the Select and Focus phase are deep uncertainties crucial for Stardust's execution such as advanced materials in space environments, the behaviour of robotic systems to manufacture at scales, and risks posed by certification schedules and supply chain volatility. These risks and deep uncertainties highly matter for delivering success and hence should have prime focus from a strategic point of view.

Why?

The phase is instrumental at improving decision-making efficiency by focusing in selected areas where more clarity would provide the greatest benefit. Instead of developing exhaustive risk inventories, planning is directly focused on uncertainties that are high-impact but under researched. The targeted approach advances the overall objective of the project to produce space components rapidly, cost effectively and sustainably.

Where?

The process is unfolded at Stellar Innovations' advanced manufacturing facilities but has to also account for interaction with a handful of external partners located remotely and separately institution-wise, such as NASA and ESA. Regulatory complexity arises from these external interfaces rendering firmer co-alignment mechanisms necessary.

When?

Following the Execution and Delivery cycle, it is applied outright at key decision-worthy points such as following prototype testing or third-party compliance reviews where the status of key uncertainties may have changed. The phase has been designed to provide iteration means, so that the planning team itself can adjust on the insights gained over time.

Who?

The key stakeholders involve internal engineering, operations, and quality assurance teams, as well as external factors such as regulators and commercial clients. Each entity views risk with a different perspective and, therefore, contributes to a more holistic treatment of uncertainties. Cross-functional engagement enables these varying assumptions to emerge early in the discussions and be resolved together.

Which Way?

The project employs a mixture of uncertainty mapping, influence diagrams, and the riskefficiency frontier way to rank learning efforts topmost. Visuals thereby come into use to help the team distinguish between uncertainties that are impactful, well-understood and those that merit further inquiries. Scenario planning methods allow structured foresight, simultaneously modelling, for example, the consequences of a six-month delay in the materials acquisition process. Consequently, continuous feedback loops keep the Select and Focus phase working in tandem with project dynamics.

How Much?

With the project's investment in costly materials, the expense of delayed regulation and the strategic expenses of postponed delivery, prioritization effectively needs to happen critically. This stage enables resource allocation to be guided by opportunity efficiency ensuring maximum return on efforts seeking clarity.

In brief, modifying the Select and Focus stage with the 7Ws results in Project Stardust's planning process being flexible, strategic, and Enlightened Planning aligned. This approach recognizes that in highly innovative and uncertain situations, clarity is not assumed—it is created through systematic stakeholder dialogue, iterative modelling, and continuous reassessment.

5. Overcoming Challenges in Adaptation of Select and Focus for Stardust

While the Select and Focus phase offers a strategic advantage in resolving execution-stage uncertainties, its implementation on a risky, high intricate project like Project Stardust will not be challenge free. To ensure the tailored application of this phase to be both practicable and

effective, it is needed to anticipate and eliminate the barriers that will be faced. The following are main implementation hurdles and associated solutions in Enlightened Planning principles:

Challenge	Explanation	Solution for Stardust
Cognitive Bias	Project teams may focus on familiar or historically prominent risks and overlooking novel uncertainties.	Conduct Delphi workshops involving outsider experts (e.g., regulatory consultants, material scientists) to challenge assumptions and expand the scope.
Ambiguity in Early Data	Certain uncertainties, such as performance of novel materials in space, have insufficient quantifiable data.	Employ exploratory scenario planning and narrative-driven logs in place of static probability-impact matrices.
Interdepartmental Misalignment	Different functional teams (compliance, operations, engineering) may hold conflicting perceptions of the most critical uncertainties.	Utilize Multi-Criteria Decision Analysis (MCDA) in cross-functional workshops to unify diverse perspectives and risk rank.
Pressure to Expedite Execution	Commercial or internal deadlines could drive early commitment ahead of clarity.	Plan execution milestones as learning milestones—such that significant uncertainties are addressed before progressing.

Dynamic Uncertainty	New and emerging uncertainties may arise as the project progresses towards execution.	Frequent Select and Focus reviews and live dashboards of uncertainty need to be built in to regularly refresh risk priorities.
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6. Conclusion

The Select and Focus phase of the SPP enables Project Stardust to approach the Execution and Delivery stage not with rigid predictions, but with informed priorities and adaptive readiness. By identifying key uncertainties—such as material behaviour, system reliability, and stakeholder expectations—and structuring project governance around learning and clarity, the team can safeguard performance while embracing innovation.

It is exactly what differs from traditional risk registers, which serve to remind and overload teams with exhaustive lists and vague mitigation plans. Enlightened Planning replaces strategic attention and learning agility for a project at the frontier of aerospace manufacturing, which possibly is a determining factor between pioneering successes or costly missteps.

Part 2: Shaping Base Plans Using Models in Project Stardust

1. Introduction to the Shape Base Plans Phase

Following the Select and Focus stage of the SPP, Base Plans Shaping Using Models is the second in the Enlightened Planning sequence. This step translates critical uncertainties into flexible, testable project plans. Rather than schedules but working hypotheses, base plans have structured flexibility. As Brito and Stafford note, it is to reflect the changing reality of the project—not control it—by means of scenario modelling and probabilistic assumptions. These plans make timing, sequencing, resource, and risk decisions easier when there is high uncertainty.

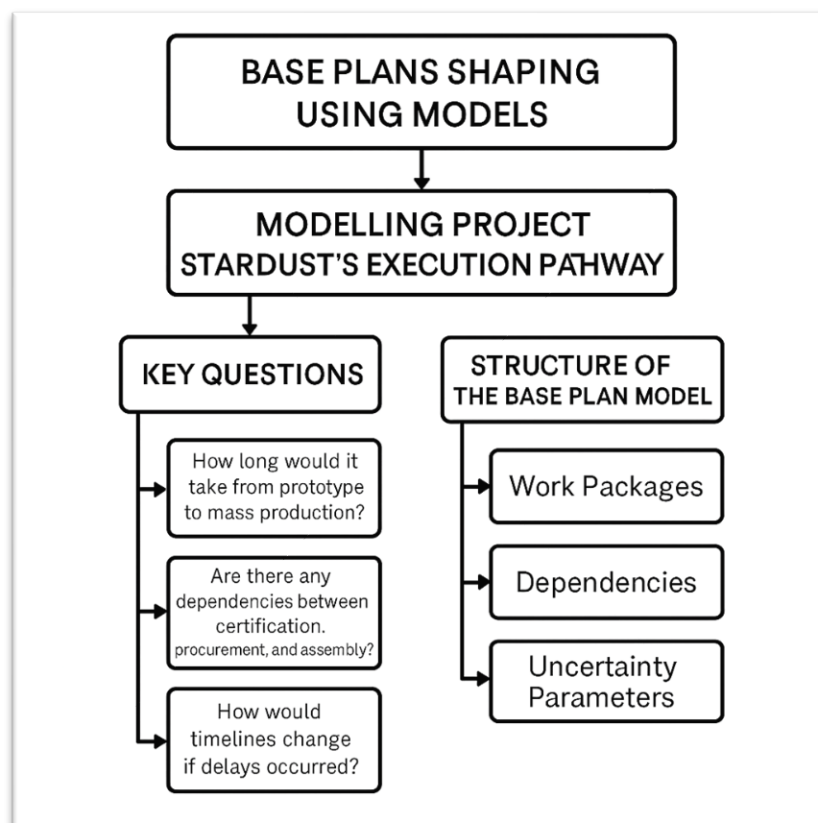


Fig 2. Shaping Base Plans Using Models in Project Stardust Flowchart

2. Modelling Project Stardust's Execution Pathway

In sculpting a viable plan for **Project Stardust**, the team must construct a model that simulates the process of **Execution and Delivery** under various plausible scenarios. The model should serve to respond to key questions:

- How long would it take for the whole scale-up efforts from prototype to mass production?
- Are there any dependencies between certification, procurement, and assembly?
- How would timelines change if delays were witnessed on raw material supply or stakeholder approval?

3. Structure of the Base Plan Model

3.1 Task Breakdown

The project consists of eight dependent and linear activities, each with a specific duration and role in the workflow. These activities are: Material Inspection (21 weeks), 3D Printing Setup (12 weeks), Operator Training (10 weeks), Prototype Build (17 weeks), Compliance Review (25 weeks), Mass Production (10 weeks), Shipment Documenting (7 weeks), Final Dispatch (2 weeks). They contribute to the overall flow of the project and their sequencing and timing depend on the relationship.

3.2 Time Parameters and Slack Analysis

For construction of a feasible project schedule, all activities were analysed on a time basis like Early Start (ES), Early Finish (EF), Late Start (LS), and Late Finish (LF). Techniques of forward and backward pass were adopted to calculate the values. The difference between the early and late times determines the slack which is an indicator of a range of free times to execute a task without affecting the end date of a project. Zero-slack tasks are considered to be critical

because any slack in doing them will have a direct effect on the overall project schedule. The critical path is established by the following activities: $A \rightarrow B \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H$ under this model. These activities collectively establish the shortest project duration of 94 weeks. Operator Training is the only non-critical task having a slack of 2 weeks; therefore, a short delay will not impact the delivery of the last project.

ID	Task	Duration	ES	EF	LS	LF	Slack
A	Material Inspection	21	0	21	0	21	0
B	3D Printing Setup	12	21	33	21	33	0
C	Operator Training	10	21	30	23	33	2
D	Prototype Build	17	33	50	33	50	0
E	Compliance Review	25	50	75	50	75	0
F	Mass Production	10	75	85	75	85	0
G	Shipment Documenting	7	85	92	85	92	0
H	Final Dispatch	2	92	94	92	94	0

Fig 3. Activity Node Values

3.3 Network Diagram

The order of tasks and activity connectivity in the project are graphically depicted in a network diagram. The diagram logically defines the order of tasks distinctly from Material Inspection, splitting into two concurrent paths: one to 3D Printing Setup (Task B) and another to Operator Training (Task C). The two converge at Prototype Build (Task D), whereas linearly there passes through all the rest of the steps until Final Dispatch (Task H). The diagram also indicates the critical path, which provides an immediate reference point for project managers to monitor time-sensitive activities.

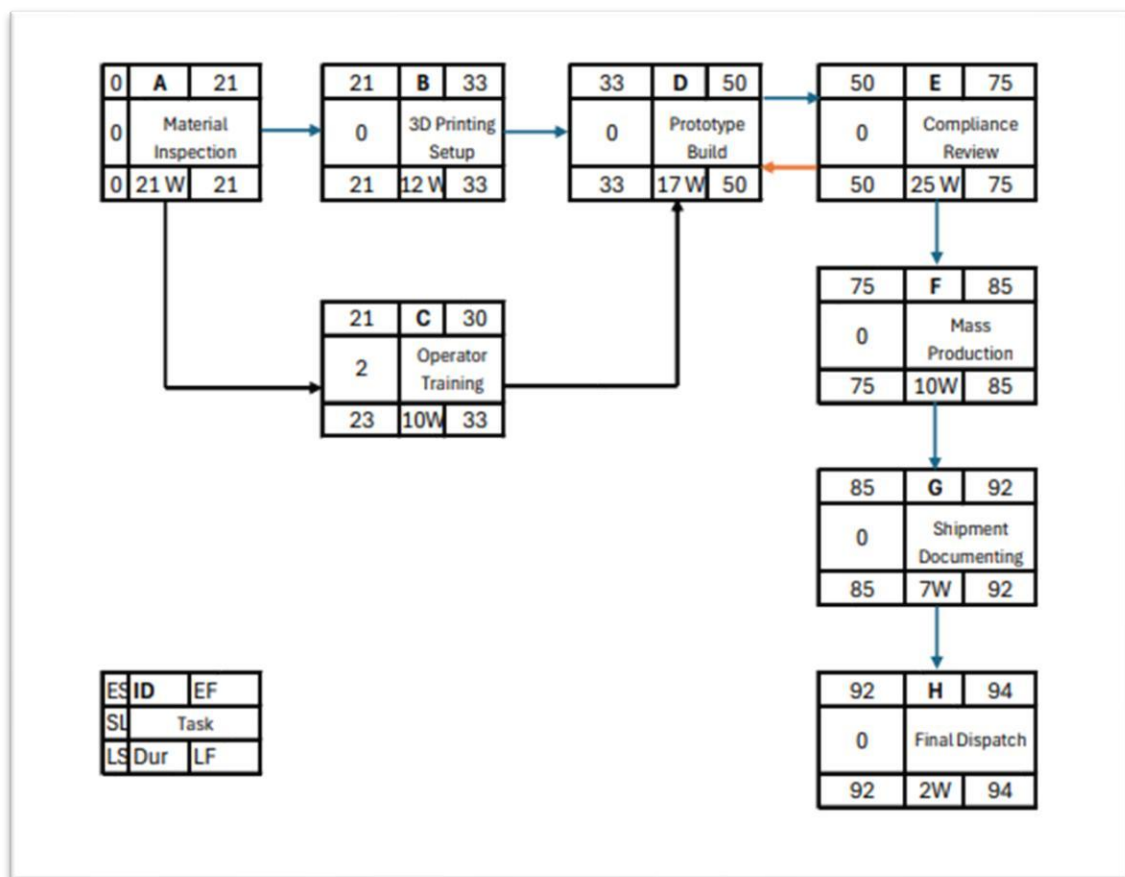


Fig 4. Activity Node Diagram

3.4 Gantt Chart (Stock-Flow Representation)

The Gantt chart shows the duration and position of each task in the project schedule graphically. It highlights overlapping tasks B and C, which occur together from Week 21 to Week 30, indicating the possibility of efficient allocation of resources. The project runs from Week 0 to Week 94, and the critical tasks are laid out in sequence and without interruption. The graphic allows us to visualize the tempo of the project and its interdependencies.

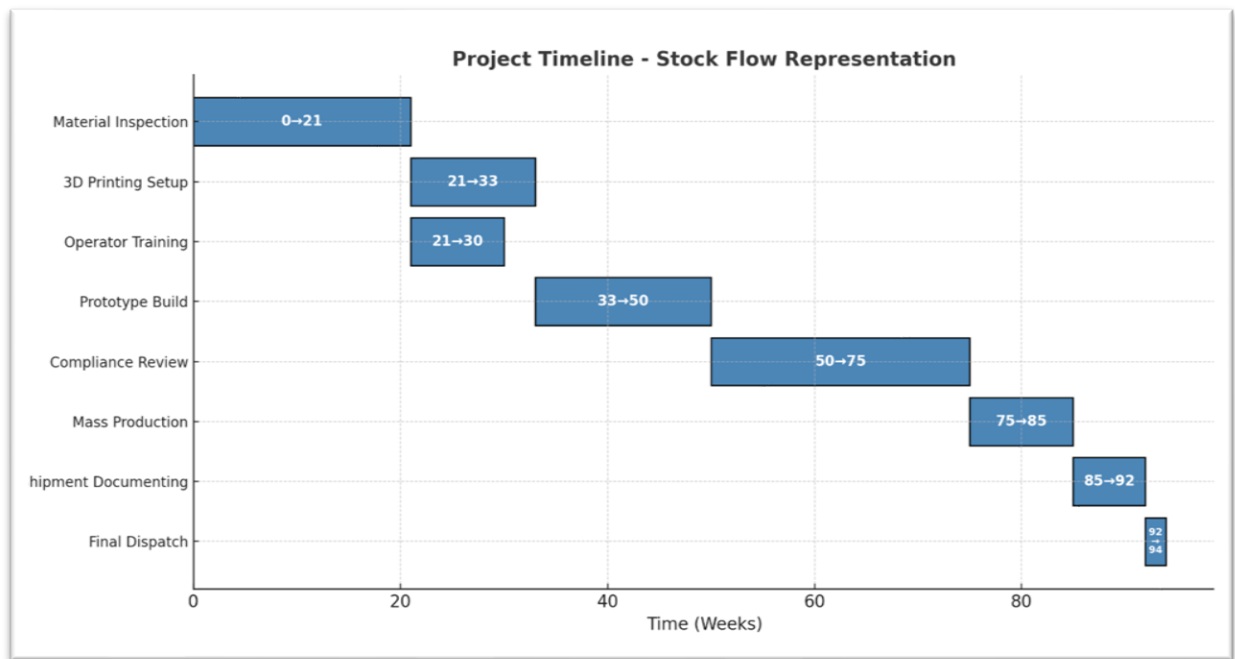


Fig 5. Gantt Chart

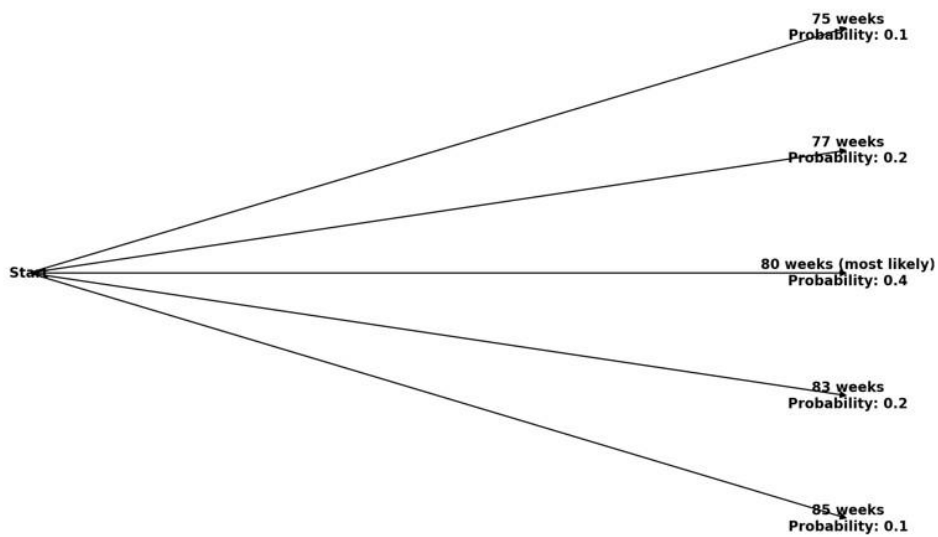


Fig 6. Probability Tree for TASK F for
10 weeks with most likely duration

80 weeks (P = 0.4)

4. Incorporating Uncertainty in the Model

1. Event Uncertainty

R1: Material space performance (Task A)

R7: Test data interpretation uncertainty (Task A)

Response: Early testing, ESA/NASA check, irreversible design freeze.

2. Inherent Variability

Relates to differing timescales because of variability in natural processes.

R4: Rare-earth supply instability (Task E)

Response: Establish stock buffers, diversify sources of supply to manage variable lead times.

3. Systemic Uncertainty

Internal project interdependencies and feedback-based uncertainty.

R2: Automated manufacturing reliability (Tasks B, C)

Response: Use simulations, Monte Carlo analysis for failure modes.

4. Ambiguity Uncertainty

Due to unclear or changing requirements and expectations.

R5: Misalignment of stakeholders (Tasks D, I)

Response: Workshops, feedback loops, MCDA for prioritisation.

5. Capability-Culture Uncertainty

Linked with organisational readiness and pressure-based decision-making.

R8: Pressure to accelerate schedules (Tasks C, E)

Action: Implement clarity thresholds, align progress with risk awareness.

Risk ID	Risk Description	Category	Impact	Likelihood	Priority	Response Strategy	Associated Task(s)
R1	Material performance in space environments	Technical	High	Medium	High	Increase early-stage testing, collaborate with ESA/NASA for review, use uncertainty modelling	A: Material Inspection
R2	Reliability of automated manufacturing systems	Operational	High	Medium-High	High	Introduce redundant systems, run pilot simulations, use Monte Carlo analysis for failure modes	B: 3D Printing Setup, C: Prototype Build
R3	Delays in regulatory certification	Regulatory	High	Medium	High	Engage early with certification bodies, plan for phased certifications, allow time buffers	D: Compliance Review
R4	Volatility in rare-earth material supply	Supply Chain	Medium-High	High	High	Diversify suppliers, build stock reserves, monitor geopolitical developments	E: Mass Production
R5	Misalignment of stakeholder expectations	Stakeholder	High	Medium	High	Conduct early stakeholder workshops, use MCDA for shared prioritisation, create feedback loops	I: Shipment Documenting, D: Compliance Review
R6	Cognitive bias in uncertainty prioritisation	Human/Process	Medium	Medium	Medium	Apply Delphi method, involve external reviewers	All Tasks
R7	Ambiguity in early data for material testing	Technical	Medium	High	Medium-High	Use narrative logs and scenario planning, delay irreversible design freezes	A: Material Inspection
R8	Internal pressure to accelerate timelines	Strategic	Medium-High	Medium	Medium-High	Link progress to clarity thresholds, educate leadership on risk of premature commitments	C: Prototype Build, E: Mass Production

Fig 7. Risk Registers

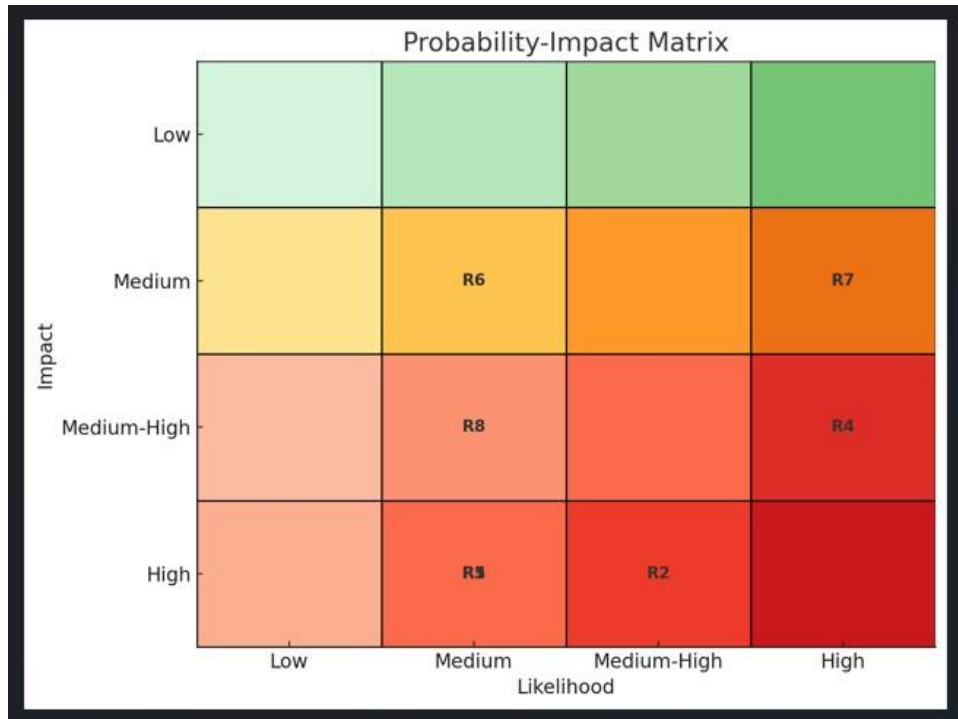


Fig 8. Probability Impact Matrix

5. Clarity and Uncertainty

In Enlightened Planning, clarity and uncertainty develop together—clarity is found by exploring uncertainty. In Project Stardust, material performance uncertainty erodes design and certification confidence. Clarity can be maximized by initial testing and ESA/NASA involvement. Modelling also reveals weak assumptions and tight buffers. Flexibility must be put into the plan if clarity is not possible.

6. Aspirational vs. Commitment Targets

Aspirational Targets

The most potential goals of the project in ideal situations, utilized for producing peak performance and inspiring stakeholders in support of the strategic direction of the project.

Total Duration: 94 weeks.

No delays or disruptions presumed in material performance, regulatory approvals, or production.

Single-path flow with deterministic scheduling.

Mass production at peak efficiency (Task F).

Assumes minimal rework or feedback loops (especially for Tasks D and E).

No supply chain critical issues, operator training delays, or organisational bottlenecks.

Used for stakeholder alignment, budgetary negotiation, and strategic communication.

Commitment Targets

Commitment targets are a realistic, risk-based schedule—used internally to manage delivery, with tolerance for uncertainties.

Extended Duration Range: 94–105+ weeks, incorporating modelled uncertainties.

Feedthrough loops from E to D may insert buffer.

Scenario-hypothesis slack inserted per time and sequence per Enlightened Planning.

Buffer to Task C slacks due to resource/operator limitation.

Response capabilities defined for determined key risks (e.g., R1–R5) with buffers, substitute suppliers.

Element	Aspirational Target	Commitment Target
Duration	94 weeks (ideal flow)	94–105+ weeks (buffered & flexible)
Risk Consideration	Minimal	High (R1–R8 modeled with mitigation)
Rework Loops	Not included	Included (e.g., Task E → D)
Flexibility	Rigid	Structured (feedback loops, buffers)
Resource Planning	Optimistic	Realistic with slack (Task C)
Purpose	Vision, motivation, stakeholder alignment	Execution, adaptation, monitoring

Fig 9. Aspirational vs. Commitment Targets

7. Using the Model for Decision-Making and Iteration

The Stardust foundation plan model should not be static but rather dynamic. It should be updated at key points of decision (e.g., following material validation) include sensitivity analysis to highlight significant uncertainties, and be communicated between teams to align. For instance, if a 10-day certification slip leads to a 30-day launch slip, teams would apply parallel testing, re-sequence the tasks, or bargain phased certification. This approach is characteristic of the Enlightened Planning principle that separation occurs through continuous learning and accommodation.

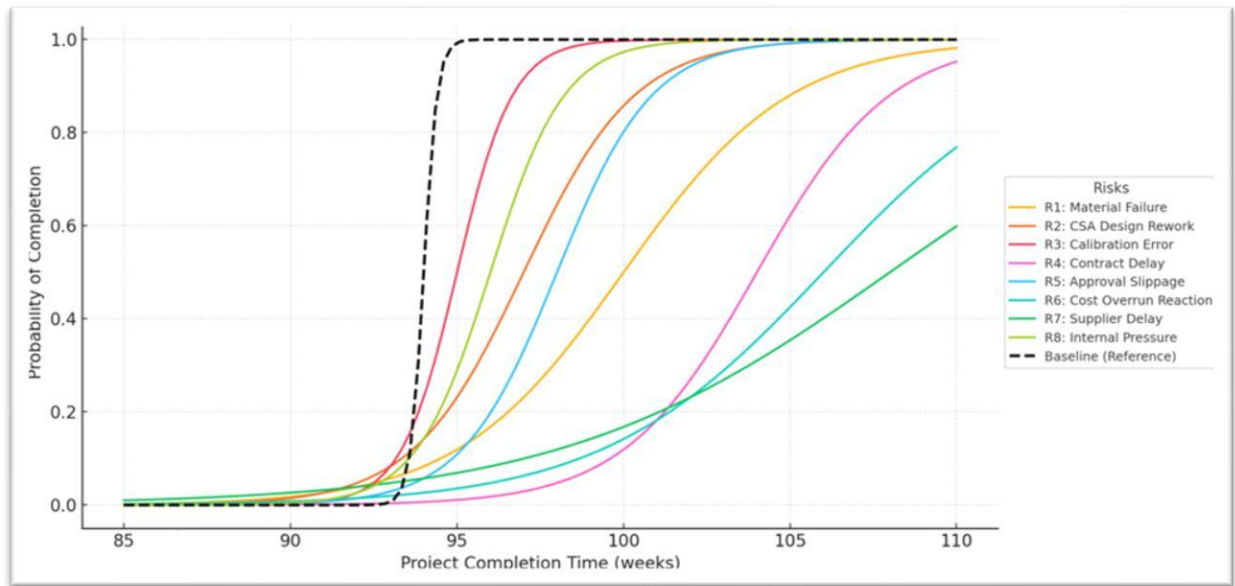


Fig 10. Risk Sensitive Analysis Diagram

8. Feedback Loops and Organizational Learning

Modelling must be a process of ongoing learning. Test outcome and review feedback must be used to develop model refinements, continually building in clarity over the lifecycle. This reinforces the Enlightened Planning principle that clarity is something acquired, rather than assumed.

9. Conclusion

Base plan modelling is also important to Project Stardust in order to manage execution-stage uncertainty. It is rather more of a decision-support resource which determines learning, timing, and commitments.

Part 3: Limitations of the PMBOK® Guide in the Context of Project

Stardust

1. Introduction

The PMI PMBOK Guide gives a structured project management methodology suitable for stable, clearly defined projects. However, it is not sufficient for technology-based highuncertainty projects like Project Stardust, involving bleeding-edge robotics, testing materials, and unstable supply chains. This report compares the PMBOK method to the more adaptive and uncertainty-sensitive Enlightened Planning (EP) framework, listing six significant deficiencies of PMBOK and how EP offers better alternatives.

Feature	PMBOK Approach	Enlightened Planning Approach
1. Treatment of Uncertainty	Focuses on measurable risks; ambiguity is under-addressed.	Explicitly distinguishes between risk and uncertainty; treats ambiguity as actionable.
2. Use of Modelling	Relies on static tools with limited modelling emphasis.	Prioritizes scenario-based modelling and simulation as central to planning.
3. Clarity as a Planning Variable	Assumes clarity grows linearly and can be baseline-locked early.	Treats clarity as a dynamic, earned asset that evolves with learning and iteration.
4. Stakeholder Engagement	Primarily transactional; limited strategic input.	Actively involves stakeholders in assumption testing, scenario exploration, and alignment.
5. Opportunity Management	Minimal focus on positive risks or aspirational strategies.	Embraces opportunity efficiency and stretch goals to drive performance.
6. Adaptability to Change	Prefers rigid baselines; change is treated as deviation.	Encourages iterative planning and real-time updates based on evolving clarity.

Fig 11. PMBOK V/S EP Approach

2. Overemphasis on Process and Control

PMBOK Guide is a predictive, linear, process-based model of risk management based on predictability and quantifiable risks. Effective in stable conditions but not in an emergent one like in Project Stardust, which has its profound, unclear risks like new materials, robot malfunctions, and changing desires from stakeholders. Enlightened Planning, on the other hand, can accommodate iterative modelling and adaptive appraisal and thus better equipped to cope with deep uncertainty.

3. Lack of Distinction Between Risk and Uncertainty

Theoretical shortcoming of the PMBOK Guide is that it does not form a clear demarcation between risk (quantifiable) and uncertainty (unquantifiable). This may lead to the improper application of quantitative methods where qualitative analysis and checking of assumptions are necessary.

All of Project Stardust's key issues that entail geopolitical impacts on international alliances or changing values of space agency stakeholders neatly fall into the realm of uncertainty, not quantifiable risk. Enlightened Planning proposes a formal separation of these categories and advises looking for clarity techniques like stakeholder workshops, structured discussion, and successive assumption tests. This enables project leaders to be in a position to be able to determine where learning, not control, is necessary—a critical competence in high-innovation environments.

4. Inflexibility to Developing Clarity

The PMBOK model assumes that project clarity such as scope, cost, and schedule—

continuously develops in a positive direction over time. Therefore, it supports initial baselining of project parameters and considers deviations as failures in performance.

However, in application, clarity inside Project Stardust emerges unevenly. New research, stakeholder feedback, or technological surprises easily alter priorities or reframe purposes during implementation. The inflexibility of PMBOK methodology inhibits adaptability in those circumstances. Enlightened Planning fosters changing clarity through tools such as phased commitment, clarity-tracked dashboards, and learning gates based upon milestones. Those tools facilitate adaptive adjustment of plans by project teams without being fined for rational directional adjustments.

5. Weak Support for Opportunity Efficiency

For Project Stardust, synergies such as early regulatory clearances, AI-based robot breakthroughs, or commonalities with other space flights could add enormous value realisation. Enlightened Planning prevents this by applying opportunity efficiency frontiers—enabling teams to propel performance aspirations with risk sensitivity preserved. Through the balance between inspirational thinking and disciplined analysis, EP enables more aspirational and value-driven planning.

6. Stakeholder Engagement Treated as Peripheral

In a project like Stardust, with organizations such as NASA, ESA, and diversified vendors, the engagement of stakeholders is essential to decide on project success. Expectation mismatch or strategic intent mismatch may result in execution going astray. EP incorporates stakeholders at the strategic level through Select and Focus workshops, assumption alignment exercises, and scenario planning combined. These processes exchange richer knowledge and build shared ownership of uncertainties and yet-to-be-made decisions.

7. Insufficient Emphasis on Modelling and Iteration

PMBOK has minimal guidance on the use of modelling tools like Monte Carlo simulation, system dynamics, or uncertainty mapping—key to addressing complexity in Project Stardust. EP puts modelling at the centre of planning, enabling iterative learning and decision-making under uncertainty.

8. Conclusion

While PMBOK is suitable for stable projects, it lacks the degree of flexibility required in highuncertainty projects. Enlightened Planning bridges this deficiency by making the difference between uncertainty and risk, thus enabling clarity-based planning and ongoing modelling and co-operation. With regard to breakthrough aerospace projects, EP is a better, adaptable planning tool.

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